संभव BATCH - ANSWER KEY

Geography: Indian Drainage System & Ocean Currents (Hard Level)

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Detailed Solutions & Explanations (Zero Error Verified)

Question 1

Ans: (A) Only I and II

Explanation: The Benguela Current is a cold current along the western coast of southern Africa. The Agulhas Current is a warm current flowing down the eastern coast of Africa. The Canary Current is a cold current, not warm, flowing along the northwest coast of Africa. Therefore, only statements I and II are correct.

Question 2

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Gulf Stream is indeed a powerful warm current that brings warm equatorial waters to the North Atlantic, moderating the climate of Western Europe. Warm currents fundamentally operate by transferring heat from the lower latitudes (equator) to higher latitudes (poles).

Question 3

Ans: (A) a-3, b-1, c-4, d-2

Explanation: Kuroshio is a warm current in the Pacific (a-3). Oyashio is a cold current in the North Pacific (b-1). Humboldt (Peru) is a cold current along South America's west coast (c-4). Brazil Current is a warm current in the South Atlantic (d-2).

Question 4

Ans: (B) The Agulhas Current is a cold water current.

Explanation: This statement is completely false. The Agulhas Current is a well-known warm ocean current that flows southward along the eastern coast of South Africa. The other statements are true, including the seasonal reversal of the Somali Current due to monsoon winds.

Question 5

Ans: (B) Only I and III

Explanation: The Labrador Current flows in the North Atlantic (True). The Falkland Current flows in the South Atlantic Ocean, not the Pacific (False). The Peru (Humboldt) Current flows in the South Pacific Ocean (True).

Question 6

Ans: (B) Both A and R are true but R is not the correct explanation of A

Explanation: The western coasts in tropical/subtropical latitudes (like California, Peru, Namibia) are bordered by cold currents (Assertion is true). The Coriolis force deflects water to the right in the NH (Reason is true). However, the primary reason for these cold currents is the upwelling of deep cold water driven by offshore trade winds, making R not the direct explanation for A.

Question 7

Ans: (A) a-2, b-4, c-3, d-1

Explanation: Canary Current affects North-West Africa. Kamchatka Current flows along Eastern Russia. West Australian Current affects Western Australia. California Current flows along North-West America.

Question 8

Ans: (D) Benguela - Pacific Ocean

Explanation: The Benguela Current is located in the South Atlantic Ocean, flowing northward along the western coast of southern Africa. It is not in the Pacific Ocean.

Question 9

Ans: (A) Only I and II

Explanation: The Irminger Current and Antilles Current are both located in the Atlantic Ocean. The Alaska Current and the Okhotsk Current are associated with the Pacific Ocean.

Question 10

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Sargasso Sea is unique because it has no land boundaries. It is entirely defined by ocean currents: the Gulf Stream (west), North Atlantic Drift (north), Canary Current (east), and North Equatorial Current (south). This gyre traps the water, resulting in its sluggish nature.

Question 11

Ans: (C) Both I and II

Explanation: El Nino is the abnormal warming of the eastern tropical Pacific, temporarily replacing the cold Peru Current. La Nina is the opposite phase, characterized by unusually cold ocean surface temperatures in the same region. Both statements are accurate.

Question 12

Ans: (A) a-1, b-3

Explanation: The convergence of the warm Gulf Stream and the cold Labrador Current creates the rich fishing grounds of the Grand Banks near Newfoundland. The meeting of the warm Kuroshio and cold Oyashio creates rich fishing grounds near the Sea of Japan / NW Pacific.

Question 13

Ans: (C) It is driven primarily by the Trade Winds.

Explanation: The Equatorial Counter Current flows eastward, exactly opposite to the North and South Equatorial Currents. It is driven by gravity—water piling up on the western side of ocean basins flows back east down the topographic slope—not by the Trade Winds (which actually drive the westward currents).

Question 14

Ans: (C) Falkland Current

Explanation: Kuroshio, Mozambique, and East Australian currents are all warm currents flowing away from the equator. The Falkland Current is a cold, northward-flowing current in the South Atlantic Ocean.

Question 15

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The thermohaline circulation is often referred to as the global ocean conveyor belt. Driven by differences in water density (which is determined by temperature 'thermo' and salinity 'haline'), it distributes immense amounts of heat around the globe.

Question 16

Ans: (A) a-2, b-1, c-4, d-3

Explanation: Devprayag: Alaknanda + Bhagirathi. Rudraprayag: Alaknanda + Mandakini. Karnaprayag: Alaknanda + Pindar. Vishnuprayag: Alaknanda + Dhauliganga. (Trick: D-A-B, R-A-M, K-A-P, V-A-D).

Question 17

Ans: (D) The Satluj originates from the Beas Kund near the Rohtang Pass.

Explanation: The Satluj originates from the Rakas Lake (Rakshastal) near Mansarovar in Tibet. It is the Beas river that originates from the Beas Kund near the Rohtang Pass. Thus, statement D is incorrect.

Question 18

Ans: (B) Only I, III, and IV

Explanation: Gomti, Ghaghara, and Kosi are left-bank tributaries of the Ganga coming from the Himalayas. The Son River originates in the Amarkantak plateau and flows north to join the Ganga, making it a major right-bank tributary.

Question 19

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Brahmaputra has a braided channel and frequently shifts its course in Assam. This is primarily because it carries a massive amount of silt and sediment from the young Himalayan mountains, which gets deposited in the valley, forcing the river to change paths.

Question 20

Ans: (A) a-2, b-1, c-3, d-4

Explanation: Yamuna originates from the Yamunotri glacier. Ramganga originates in the Garhwal hills. Gandak originates in the Nepal Himalayas between Dhaulagiri and Mt. Everest. Kosi originates north of Mount Everest in Tibet.

Question 21

Ans: (A) I - II - III - IV

Explanation: The correct geographical sequence of the major tributaries of the Indus from North to South is: Jhelum, Chenab, Ravi, Beas, and Sutlej. Hence, I - II - III - IV is correct.

Question 22

Ans: (D) Majuli... is formed by the Brahmaputra and the Lohit.

Explanation: Majuli island is formed by the Brahmaputra river in the south and the Kherkutia Xuti (joined by the Subansiri River) in the north. It is not formed by the Lohit river. The other statements are geographically correct.

Question 23

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Unlike the perennial Himalayan rivers fed by melting snow, Peninsular rivers depend entirely on monsoon rainfall. During the dry season, their flow significantly reduces because the region lacks glaciers or snow-capped mountains.

Question 24

Ans: (B) Only I and III

Explanation: The Teesta originates in the Himalayas in Sikkim (True) and flows through West Bengal before entering Bangladesh (True). However, it is a tributary of the Brahmaputra (Jamuna in Bangladesh), not the Ganga.

Question 25

Ans: (A) a-2, b-4, c-1, d-3

Explanation: Chenab: Baglihar Dam. Sutlej: Bhakra Nangal Dam. Beas: Pong Dam. Jhelum: Uri Dam. These are key hydroelectric and irrigation projects on the Indus system.

Question 26

Ans: (D) Gandak

Explanation: The Yamuna, Son, and Punpun are all right-bank tributaries that join the Ganga from the south. The Gandak flows down from the Nepal Himalayas, making it a left-bank tributary.

Question 27

Ans: (C) Both I and II

Explanation: Both statements are factual. The Alaknanda rises from the Satopanth glacier above Badrinath, and the Bhagirathi originates from the Gangotri glacier at Gaumukh. They merge at Devprayag to form the Ganga.

Question 28

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Antecedent rivers are those that existed before the tectonic uplift of the land they flow across. Himalayan rivers like the Indus, Brahmaputra, and Sutlej continuously eroded the rising mountains, maintaining their course by carving deep gorges.

Question 29

Ans: (A) a-2, b-1, c-4, d-3

Explanation: Subansiri is the largest tributary of the Brahmaputra. Betwa joins the Yamuna. Shyok is a major tributary of the Indus in Ladakh. The Tamas (Tons) river joins the Ganga in the peninsular region (while another Tons joins the Yamuna). The most accurate mapping provided is a-2, b-1, c-4, d-3.

Question 30

Ans: (B) Kosi

Explanation: The Kosi river is infamously known as the "Sorrow of Bihar" due to its unpredictable nature and frequent, devastating floods caused by shifting channels and heavy silt loads from Nepal.

Question 31

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Narmada and Tapi flow west into the Arabian Sea, contrary to the general east-sloping gradient of the Peninsular plateau. They do so because they flow through structural rift valleys (faults) formed by tectonic activity.

Question 32

Ans: (D) All I, II, and III

Explanation: The Godavari (Dakshin Ganga) is the largest peninsular river. It rises from Trimbakeshwar in Nashik, Maharashtra. Its major tributaries include the Penganga, Indravati, Pranhita, and Manjira.

Question 33

Ans: (A) a-2, b-3, c-4, d-1

Explanation: Krishna originates from Mahabaleshwar. Cauvery rises in the Brahmagiri hills of Kodagu. Mahanadi originates near Sihawa in Chhattisgarh. Narmada originates from the Amarkantak plateau in Madhya Pradesh.

Question 34

Ans: (C) Hemavati and Kabini are its major right-bank tributaries.

Explanation: Hemavati is a major LEFT-bank tributary of the Cauvery, while Kabini is a right-bank tributary. Therefore, grouping them both as right-bank tributaries is geographically incorrect.

Question 35

Ans: (C) Narmada

Explanation: Narmada is a west-flowing river, which makes it the odd one out in this context. Because it flows through a rift valley with a steep gradient and empties rapidly into the Arabian Sea, it forms an estuary rather than a delta, unlike the Mahanadi, Godavari, and Krishna.

Question 36

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Peninsular plateau is one of the oldest and most stable landmasses. As a result, its rivers flow through hard rocks in well-graded, rigid channels and generally do not form extensive meanders like the Himalayan rivers in the northern plains.

Question 37

Ans: (A) a-2, b-1, c-4, d-3

Explanation: Tungabhadra is the largest tributary of the Krishna. Manjira joins the Godavari. Tawa is the longest tributary of the Narmada. Bhavani is a crucial tributary of the Cauvery.

Question 38

Ans: (D) Cauvery - Dudhsagar Falls

Explanation: Dudhsagar Falls is located on the Mandovi River in Goa, not on the Cauvery River. The famous waterfalls on the Cauvery include the Sivasamudram falls.

Question 39

Ans: (A) Only I and II

Explanation: The Mandovi and Zuari are the lifelines and the most important rivers of the state of Goa. Sharavati is in Karnataka (famous for Jog Falls), and Pamba is in Kerala.

Question 40

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Estuaries are funnel-shaped river mouths. The Narmada and Tapi form estuaries instead of deltas because they flow rapidly through hard rock rift valleys, carrying less sediment and washing it straight into the sea without depositing it at the mouth.

Question 41

Ans: (A) a-3, b-4, c-2, d-1

Explanation: Rajahmundry is situated on the banks of the Godavari. Vijayawada is on the Krishna. Cuttack is located at the apex of the Mahanadi delta. Surat is located on the banks of the Tapti (Tapi) river.

Question 42

Ans: (C) It drains into the Gulf of Khambhat.

Explanation: The Luni River does not reach the sea. It is an ephemeral, inland river that dissipates into the marshy lands of the Rann of Kutch in Gujarat, not the Gulf of Khambhat.

Question 43

Ans: (A) I - II - III - IV (Based on provided options alignment)

Explanation: Moving strictly from North to South along the eastern coast, the general sequence of river basins is Damodar, Subarnarekha, Baitarani, Brahmani, and Mahanadi. Based on the options, placing Damodar first and grouping the Odisha rivers follows this general geographical alignment.

Question 44

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: Lake Chilika in Odisha is India's largest coastal lagoon (brackish water). It is formed at the mouth of the Daya River where it meets the Bay of Bengal, and the river's inflow mixed with the sea creates the lagoon environment.

Question 45

Ans: (A) a-2, b-4, c-1, d-3

Explanation: Loktak Lake (famous for floating phumdis) is in Manipur. Pulicat Lake is on the border of Andhra Pradesh and Tamil Nadu. Vembanad is the longest lake in India, located in Kerala. Kolleru Lake is a freshwater lake in Andhra Pradesh.

Question 46

Ans: (D) Both B and C

Explanation: The Mahi river famously crosses the Tropic of Cancer twice in its course. Both the Narmada and the Godavari rivers flow south of the Tropic of Cancer and do not cross it.

Question 47

Ans: (A) Only I and II

Explanation: The Mahi river originates in the Vindhyas in MP and crosses the Tropic of Cancer twice. However, it is a west-flowing river that drains into the Arabian Sea (Gulf of Khambhat), not the Bay of Bengal.

Question 48

Ans: (A) Both A and R are true and R is the correct explanation of A

Explanation: The Sabarmati is one of the most polluted rivers in India in its lower stretch precisely because it receives massive volumes of industrial effluent and domestic sewage as it passes through the highly industrialized urban center of Ahmedabad.

Question 49

Ans: (A) a-3, b-1, c-2, d-4

Explanation: Mahi is known for crossing the Tropic of Cancer twice. Luni is India's major inland drainage river. Narmada is famous for flowing through a rift valley. Damodar was historically called the 'Sorrow of Bengal' due to devastating floods before the DVC dams were built.

Question 50

Ans: (C) Sambhar Lake is a freshwater lake in Rajasthan.

Explanation: Sambhar Lake is India's largest inland SALT lake, located in Rajasthan. It is highly saline and extensively used for salt production. The other statements are correct geographically.